

7. Limitations

One limitation of our research is that we only use a binary classification for labels. We recognize that scalar diversity can be more fine-grained than simple *yes* or *no*. Moreover, we also note that the datasets that we use for scalar diversity task are relatively small and only constrained to SAs, which is due to the sparsity of large-scale scalar reasoning datasets¹⁵. We encourage future works to undertake larger-scale collections of fine-grained datasets to evaluate models and train them to suit relevant needs.

Finally, some of our test datasets (e.g. ukWac) are published before the release of our models. LLMs may have already seen the test materials. As discussed in (La Malfa et al., 2024; Drinkall et al., 2024), evaluations of LLMs may be artifactually high because of the difficulties in ensuring that the test data are truly invisible during training. Future works can investigate this problem further.

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9. Bibliographical References

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¹⁵Some datasets containing SI triggers such as determiners and verbs, but they are also small (Ronai and Xiang, 2023b; Van Tiel et al., 2014).

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